ImageNet Classification with Deep Convolutional Neural Networks (pt2)

Michael Schatz
Oct 6, 2025

JHU EN.601.449/EN.601.649: Applied Comparative Genomics

Assignment 3 Due Wednesday Oct 1 by 11:59pm

github.com/schatzlab/appliedgenomics2025/blob/main/assignments/assignment3/README.md

Files

main

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- lectures
- policies
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- LICENSE
- README.md

Preview | Code | Blame 153 lines (96 loc) · 8.61 KB Raw

Assignment 3: BWT and Variant Calling

Assignment Date: Wednesday, September 24, 2025
Due Date: Wednesday, October 1, 2025 @ 11:59pm

Assignment Overview

In this assignment you will implement the BWT and explore the requirements for variant calling. The programming exercises can be computed in any programming language, although we recommend python (or C++, Java, or Rust). R is generally inefficient at string processing unless you take great care. See the resources at the bottom of the page for tips for the variant calling exercises.

As a reminder, any questions about the assignment should be posted to Piazza.

Question 1. BWT Encoding [20 pts]

In the language of your choice, implement a BWT encoder and encode the string below. Faster (Linear time) methods exist for computing the BWT, although for this assignment you can use the simple method based on standard sorting techniques. Your solution does *not* need to be an optimal algorithm and can use O(n^2) space and O(n^2 lg n) time.

Here is the recommended pseudo code (make sure to submit your code as well as the encoded string):

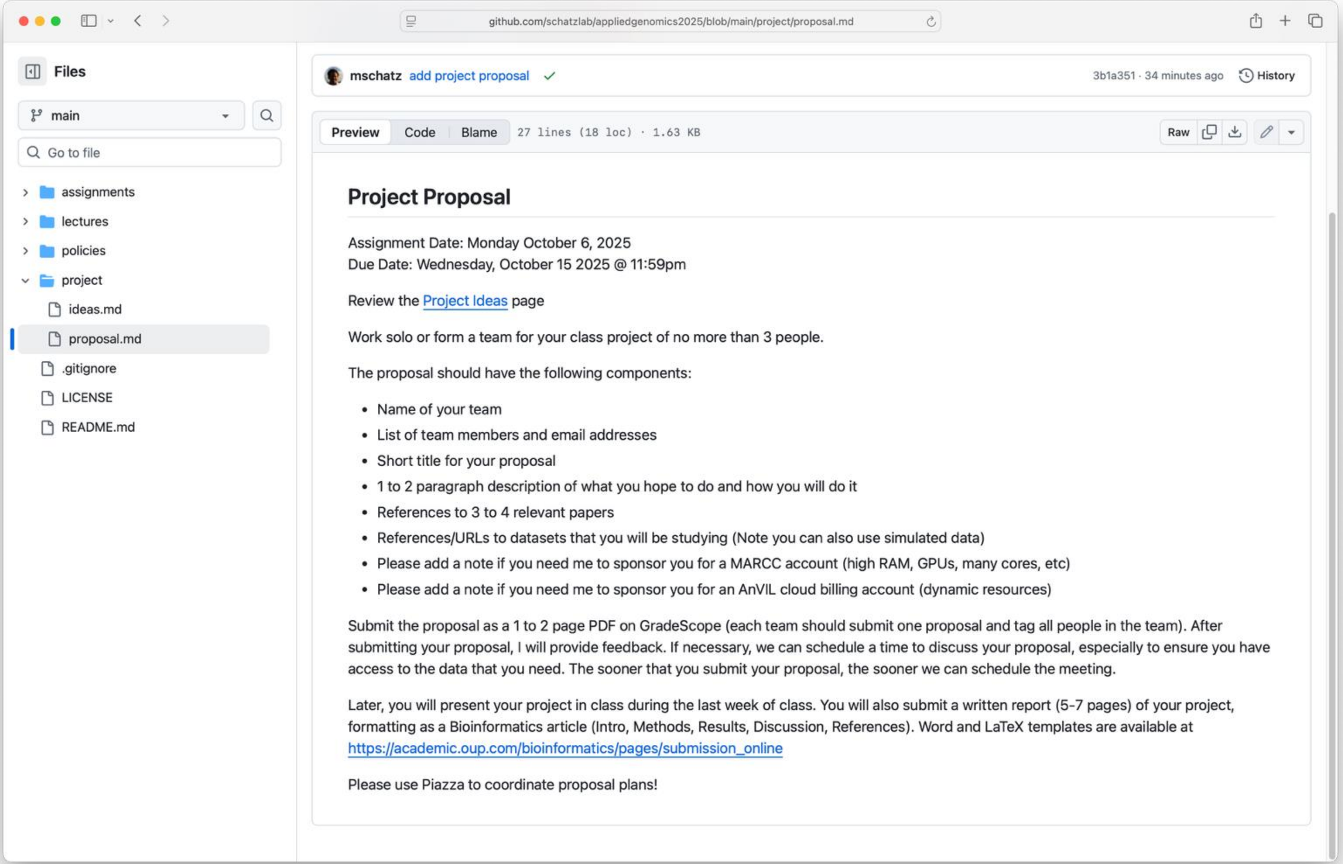
```
computeBWT(string s)
  ## add the magic end-of-string character
  s = s + "$"

  ## build up the BWM from the cyclic permutations
  ## note the ith cyclic permutation is just "s[i..n] + s[0..i]"
  rows = []
  for (i = 0; i < length(s); i++)
    rows.append(cyclic_permutation(s, i))

  ## just use the builtin sort command to sort the cyclic permutations
  sort(rows)

  ## now extract the last column
  bwt = ""
  for (i = 0; i < length(s); i++)
    bwt += substr(rows[i], length(s), 1)
```

Project Proposal Due Wednesday Oct 15 by 11:59pm



Project Proposal

Assignment Date: Monday October 6, 2025
Due Date: Wednesday, October 15 2025 @ 11:59pm

Review the [Project Ideas] page

Work solo or form a team for your class project of no more than 3 people.

The proposal should have the following components:

- Name of your team
- List of team members and email addresses
- Short title for your proposal
- 1 to 2 paragraph description of what you hope to do and how you will do it
- References to 3 to 4 relevant papers
- References/URLs to datasets that you will be studying (Note you can also use simulated data)
- Please add a note if you need me to sponsor you for a MARCC account (high RAM, GPUs, many cores, etc)
- Please add a note if you need me to sponsor you for an AnVIL cloud billing account (dynamic resources)

Submit the proposal as a 1 to 2 page PDF on GradeScope (each team should submit one proposal and tag all people in the team). After submitting your proposal, I will provide feedback. If necessary, we can schedule a time to discuss your proposal, especially to ensure you have access to the data that you need. The sooner that you submit your proposal, the sooner we can schedule the meeting.

Later, you will present your project in class during the last week of class. You will also submit a written report (5-7 pages) of your project, formatting as a Bioinformatics article (Intro, Methods, Results, Discussion, References). Word and LaTeX templates are available at https://academic.oup.com/bioinformatics/pages/submission_online

Please use Piazza to coordinate proposal plans!

AlexNet

"AlexNet"

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
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Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

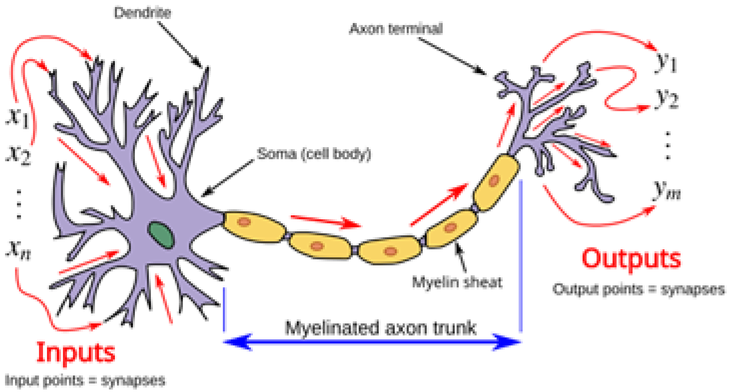
Alex Krizhevsky
U. Toronto/Google

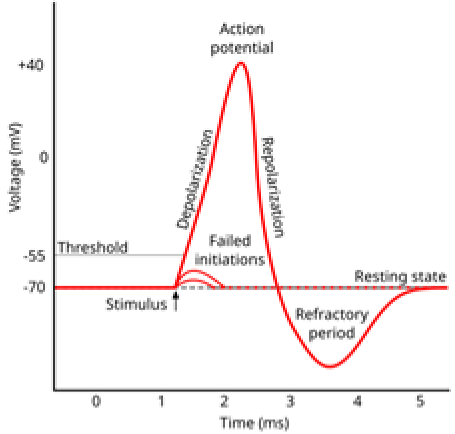
Ilya Sutskever
U. Toronto/OpenAI/
Safe Superintelligence Inc

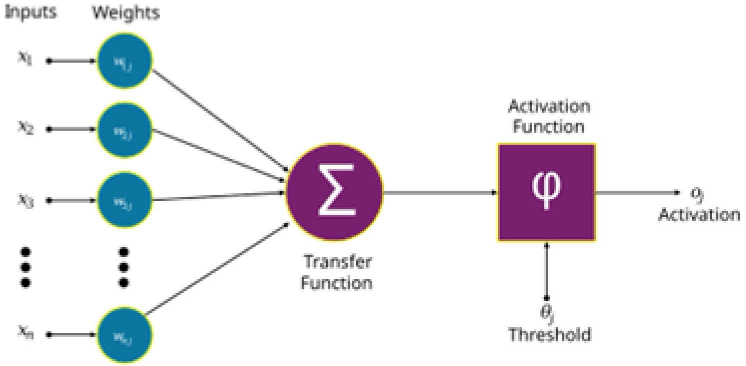
Geoffrey Hinton
U. Toronto/Vector Institute

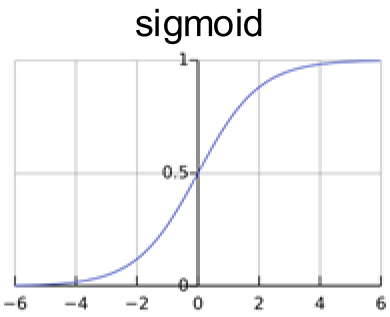
What is this?

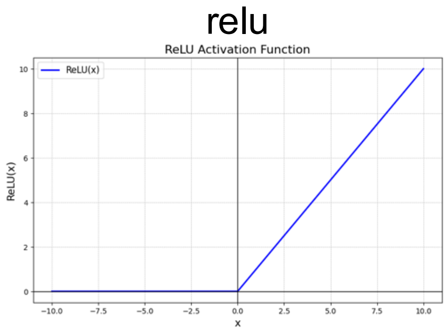
Biological vs Artificial Neurons





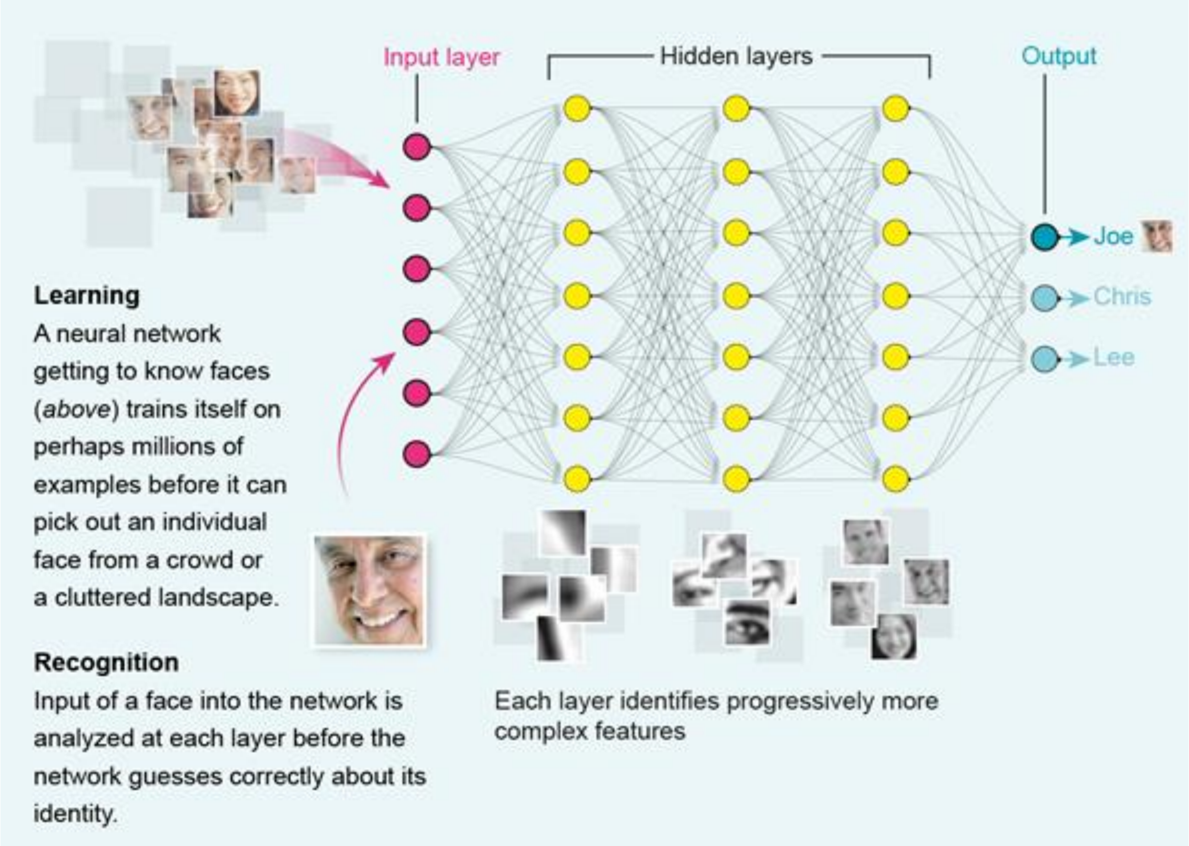






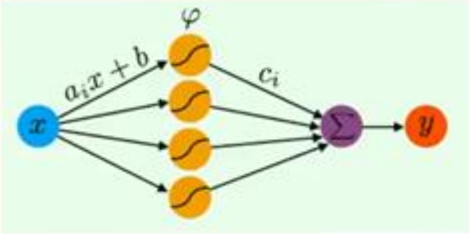
The non-linear activation function (sigmoid/relu) is needed to model non-linear relationships

Artificial Neural Networks



https://www.codingdojo.com/blog/artificial-neural-network-brain-behind-todays-smart-technology

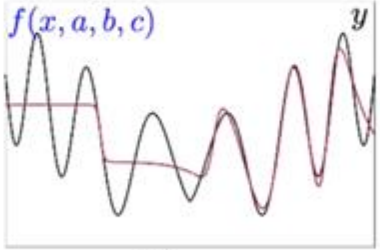
ANNs are "Universal Approximators"



1 hidden layer perceptron:

$$y \approx f(x, a, b, c) \overset{\text{def.}}{=} \sum_{i=1}^{p} c_i \varphi(a_i x + b_i)$$

$p = 6$ neurons



$p = 20$ neurons

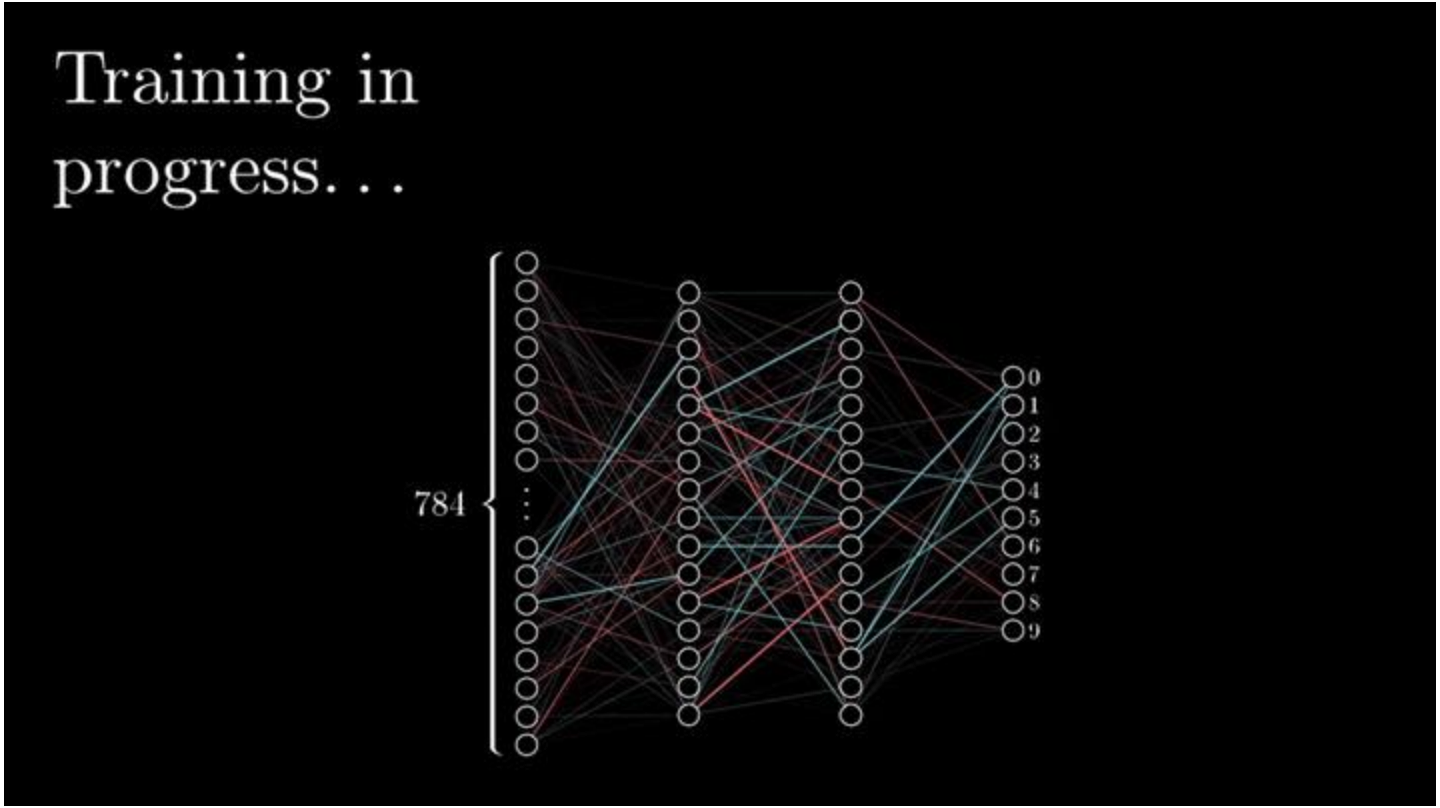
Approximation by Superpositions of a Sigmoidal Function*

G. Cybenko†

Abstract. In this paper we demonstrate that finite linear combinations of compositions of a fixed, univariate function and a set of affine functionals can uniformly approximate any continuous function of n real variables with support in the unit hypercube; only mild conditions are imposed on the univariate function. Our results settle an open question about representability in the class of single hidden layer neural networks. In particular, we show that arbitrary decision regions can be arbitrarily well approximated by continuous feedforward neural networks with only a single internal, hidden layer and any continuous sigmoidal nonlinearity. The paper discusses approximation properties of other possible types of nonlinearities that might be implemented by artificial neural networks.

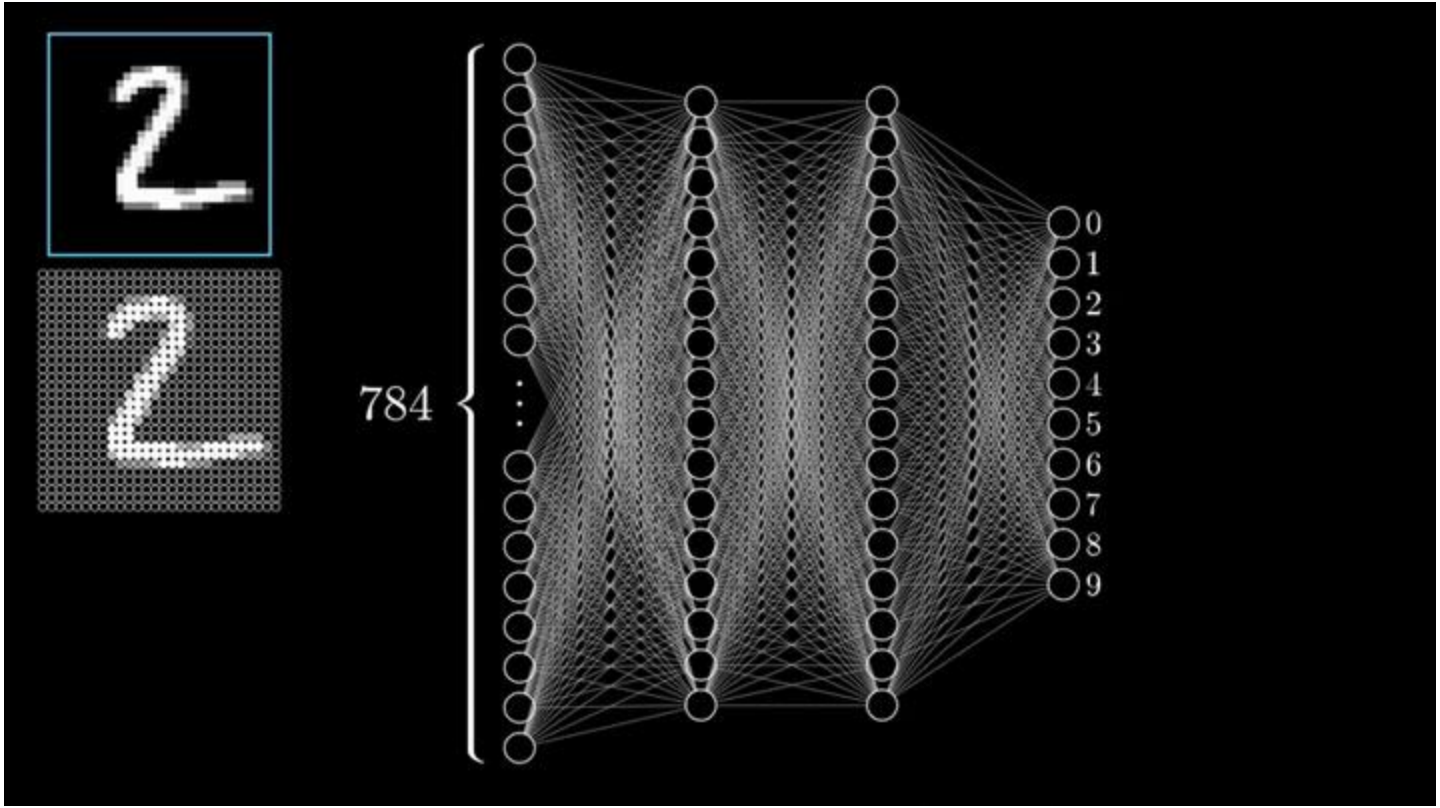
Key words. Neural networks, Approximation, Completeness.

Recap - Training



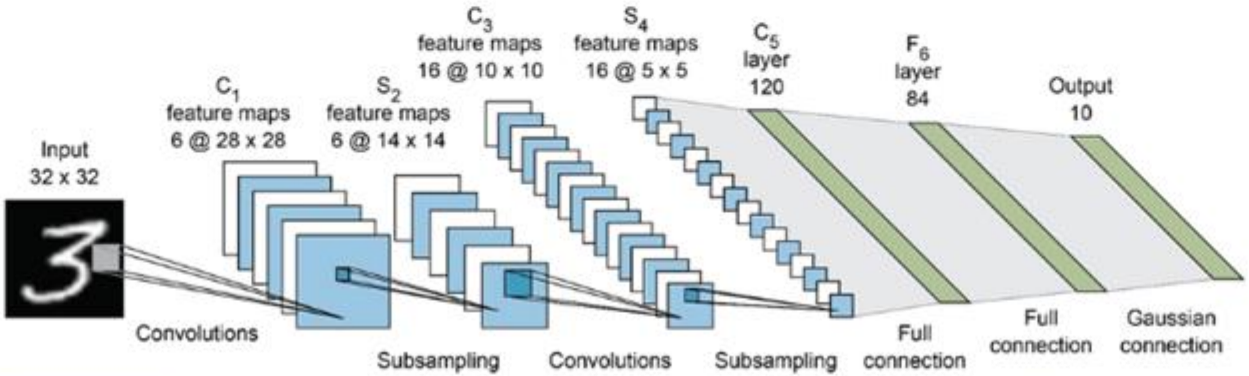
https://youtu.be/IHZwWFHWa-w

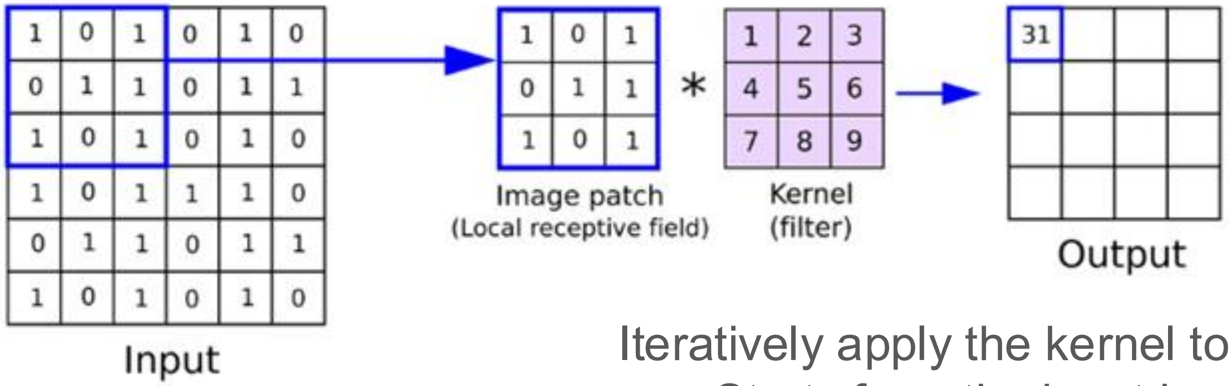
Recap - Evaluation



https://www.youtube.com/watch?v=IIg3gGewQ5U

Convolutional Neural Network





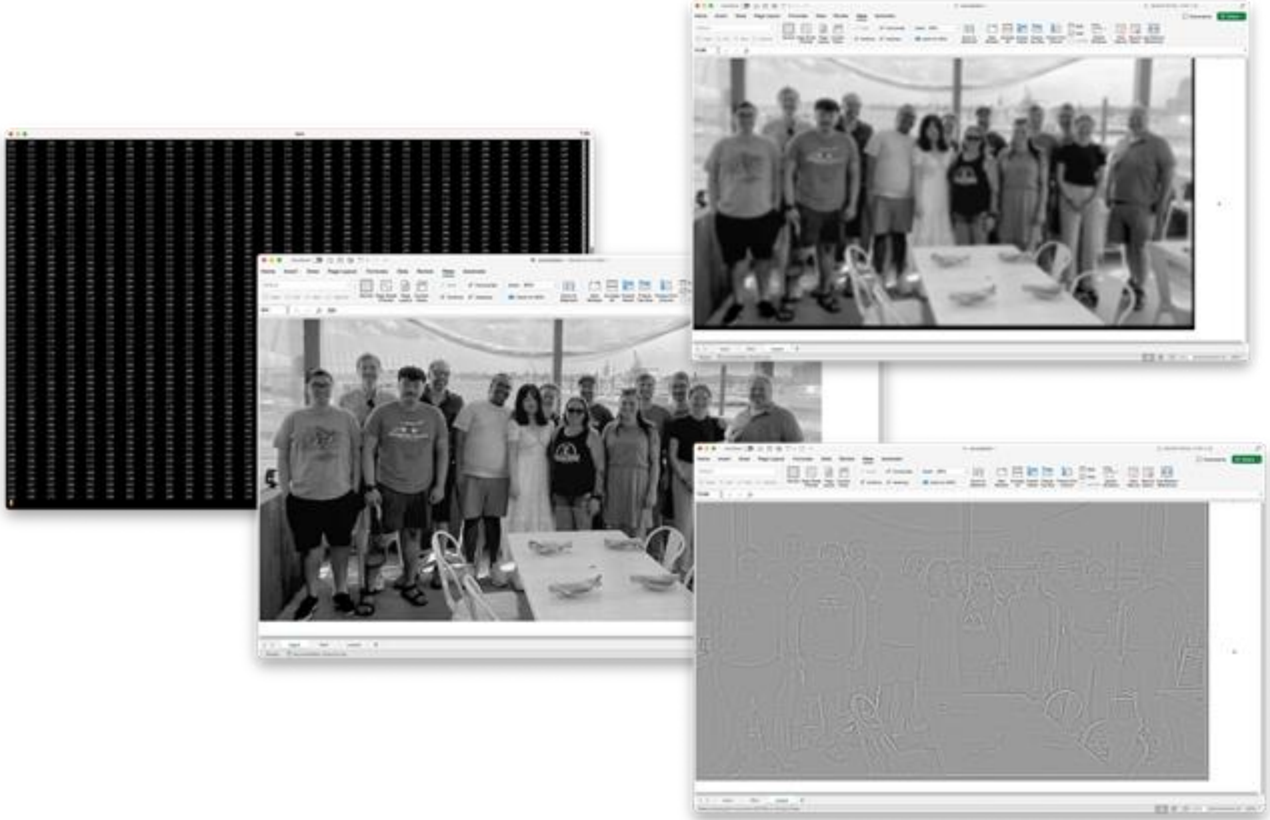
"LeNet" (Lecun et al, 1998)

Iteratively apply the kernel to extract features

- Starts from the input image, then creates more abstract derived images

Kernels

Operation	Kernel ω	Image result g(x,y)
Identity	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	
Ridge or edge detection	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$	
Sharpen	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	
Box blur (normalized)	$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	
Gaussian blur 3 × 3 (approximation)	$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	
Gaussian blur 5 × 5 (approximation)	$\frac{1}{256}\begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	
Unsharp masking 5 × 5 Based on Gaussian blur with amount as 1 and threshold as 0 (with no image mask)	$\frac{-1}{256}\begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & -476 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$	



```
= input!A1 * filter!$A$1 + input!A2 * filter!$A$2 + input!A3 * filter!$A$3 + input!A4 * filter!$A$4 + input!A5 * filter!$A$5 +
  input!B1 * filter!$B$1 + input!B2 * filter!$B$2 + input!B3 * filter!$B$3 + input!B4 * filter!$B$4 + input!B5 * filter!$B$5 +
  input!C1 * filter!$C$1 + input!C2 * filter!$C$2 + input!C3 * filter!$C$3 + input!C4 * filter!$C$4 + input!C5 * filter!$C$5 +
  input!D1 * filter!$D$1 + input!D2 * filter!$D$2 + input!D3 * filter!$D$3 + input!D4 * filter!$D$4 + input!D5 * filter!$D$5 +
  input!E1 * filter!$E$1 + input!E2 * filter!$E$2 + input!E3 * filter!$E$3 + input!E4 * filter!$E$4 + input!E5 * filter!$E$5
```

Outline

1. The Problem
2. pre-AlexNet approaches
3. How do animals & humans approach this?
4. Artificial Neural Networks
5. Convolutional Neural Networks
6. AlexNet
7. Impact

ImageNet Classification with Deep Convolutional Neural Networks

1

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3

4

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8

9

Network Architecture

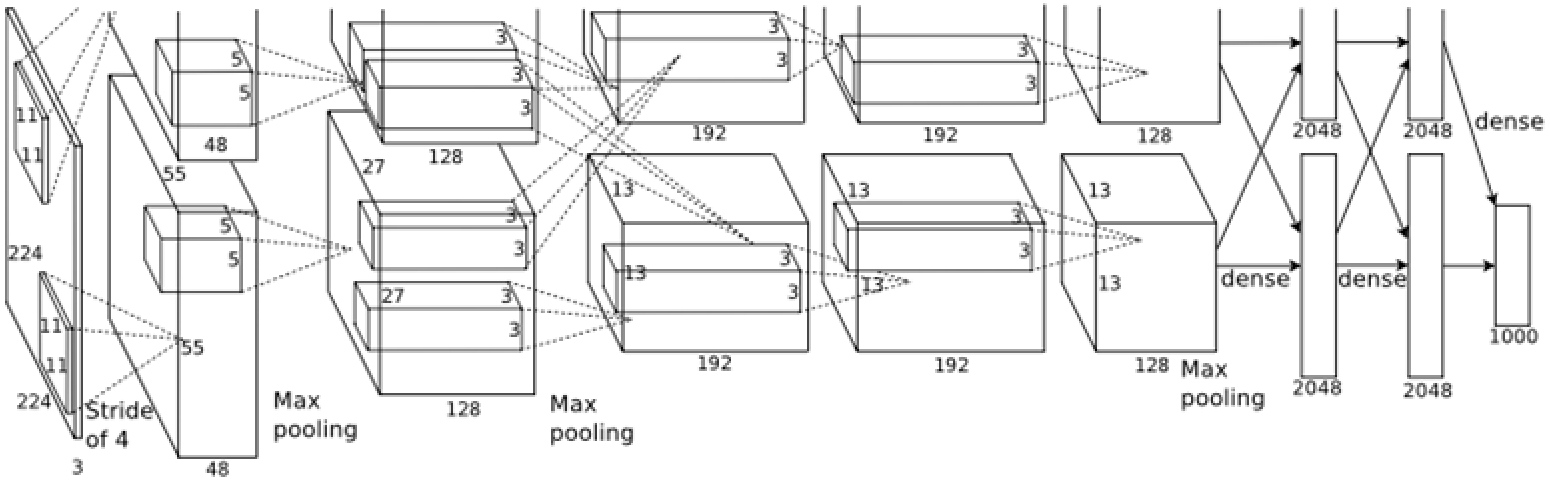
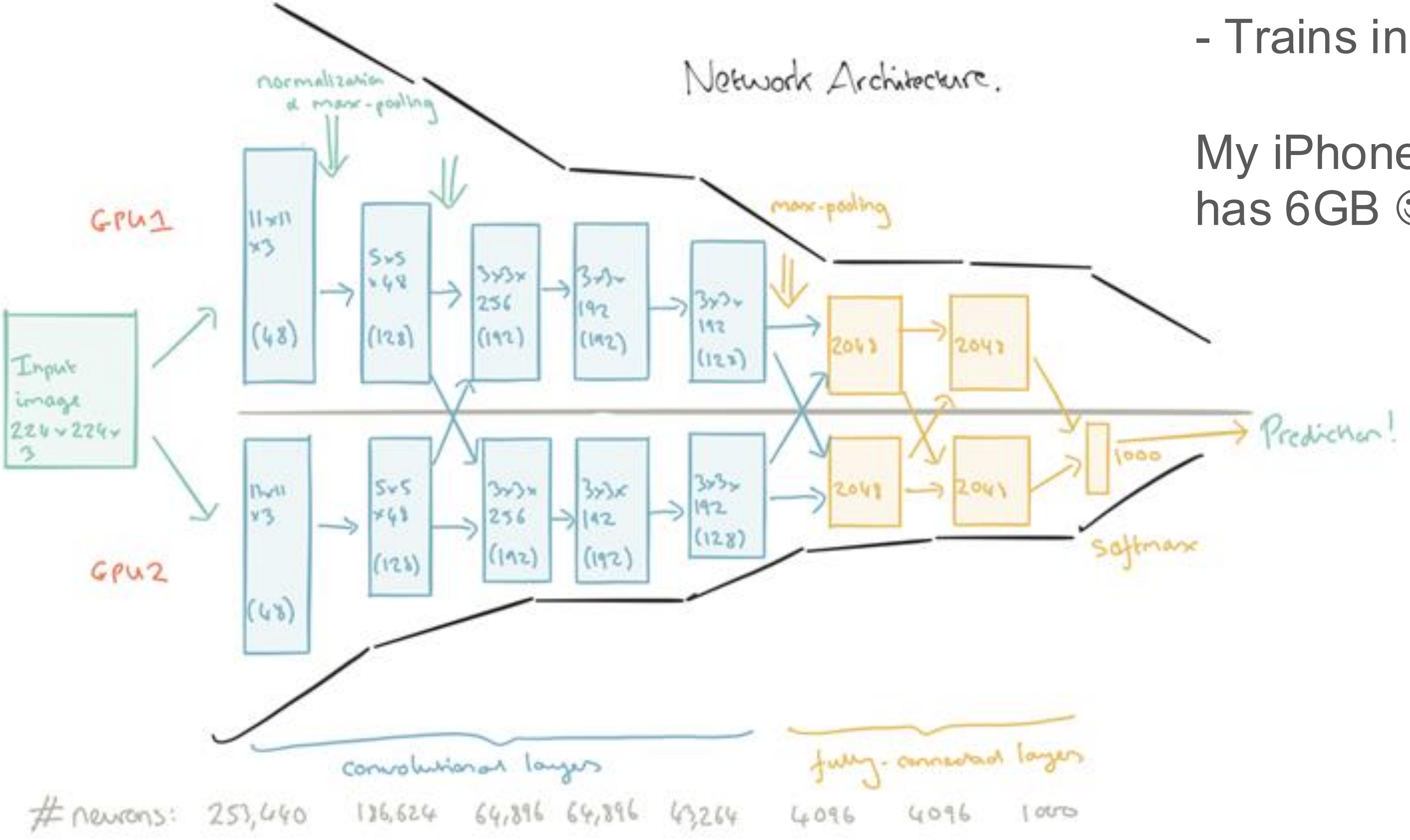


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Network Architecture (annotated)



GTX 580 GPU only has 3GB!

- Trains in ~5 days

My iPhone's GPU has 6GB ☺

https://blog.acolyer.org/2016/04/20/imagenet-classification-with-deep-convolutional-neural-networks/

Learned Convolutional Kernels

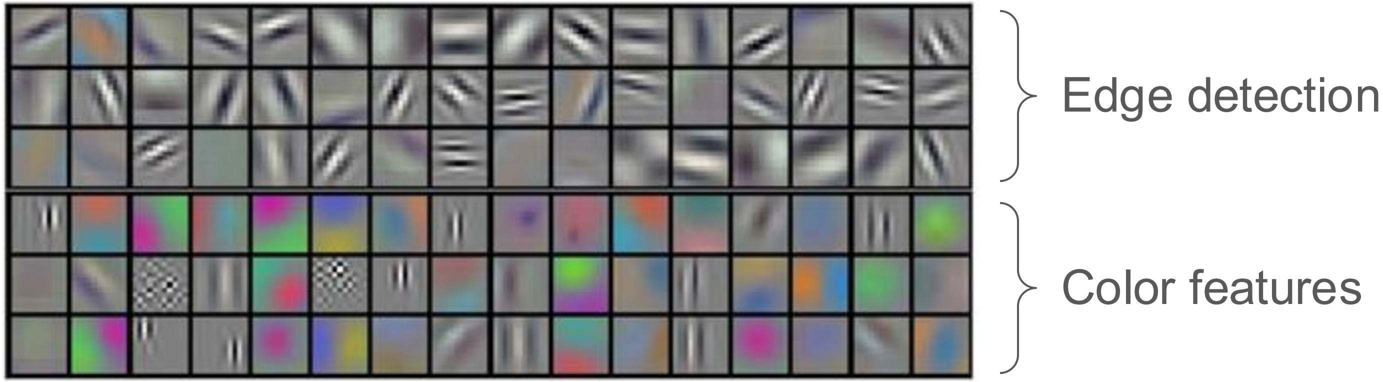
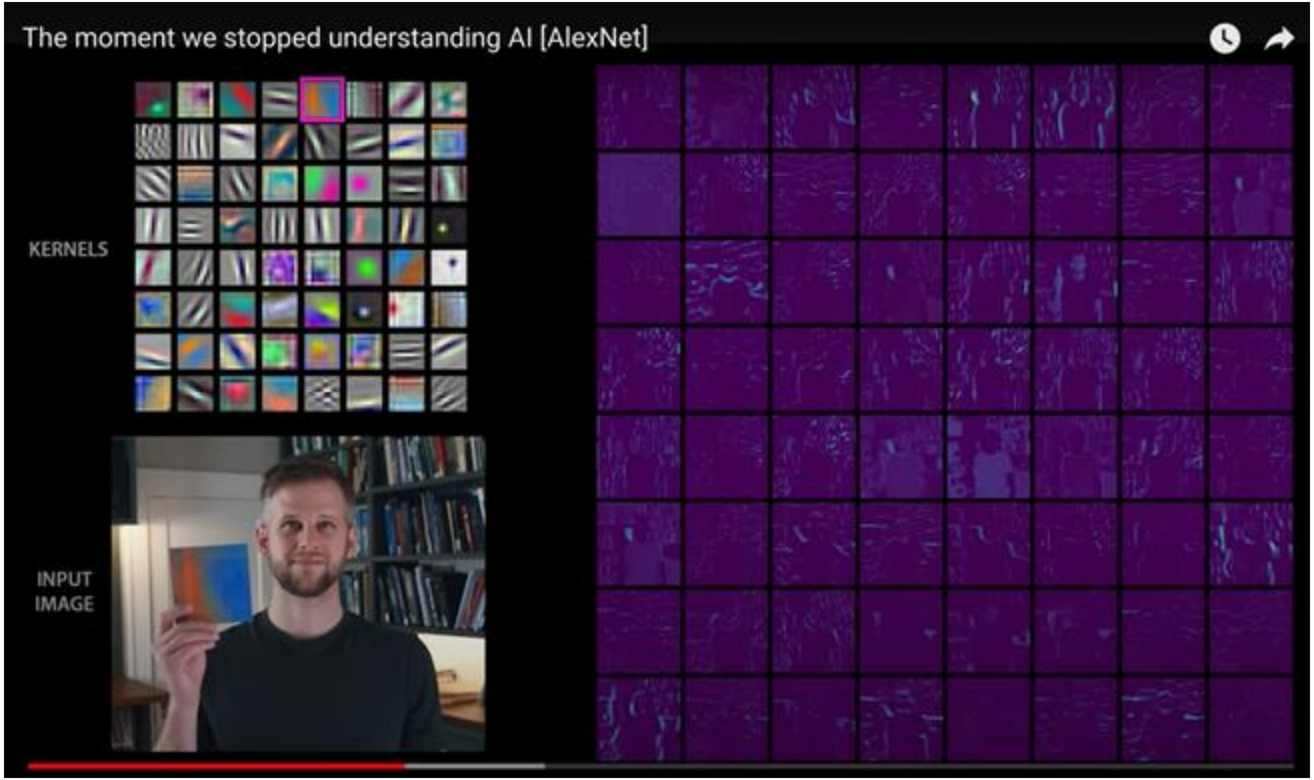


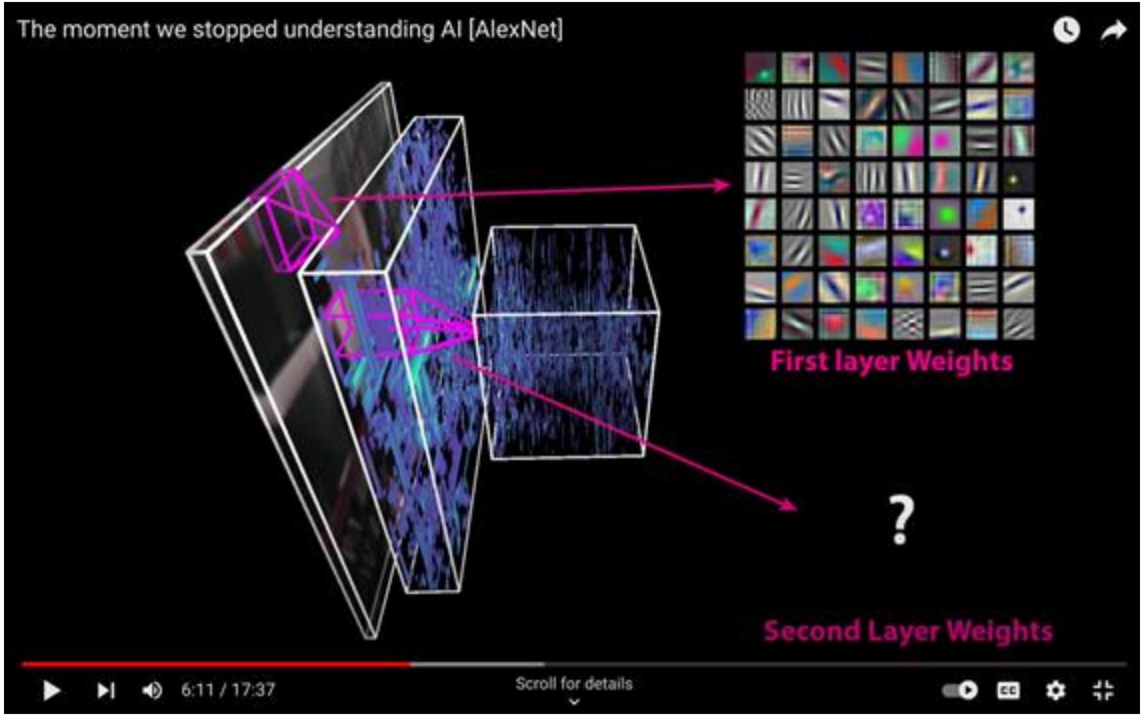
Figure 3: 96 convolutional kernels of size $11\times11\times3$ learned by the first convolutional layer on the $224\times224\times3$ input images. The top 48 kernels were learned on GPU 1 while the bottom 48 kernels were learned on GPU 2. See Section 6.1 for details.

From 1 image to 96 images: Activation Maps



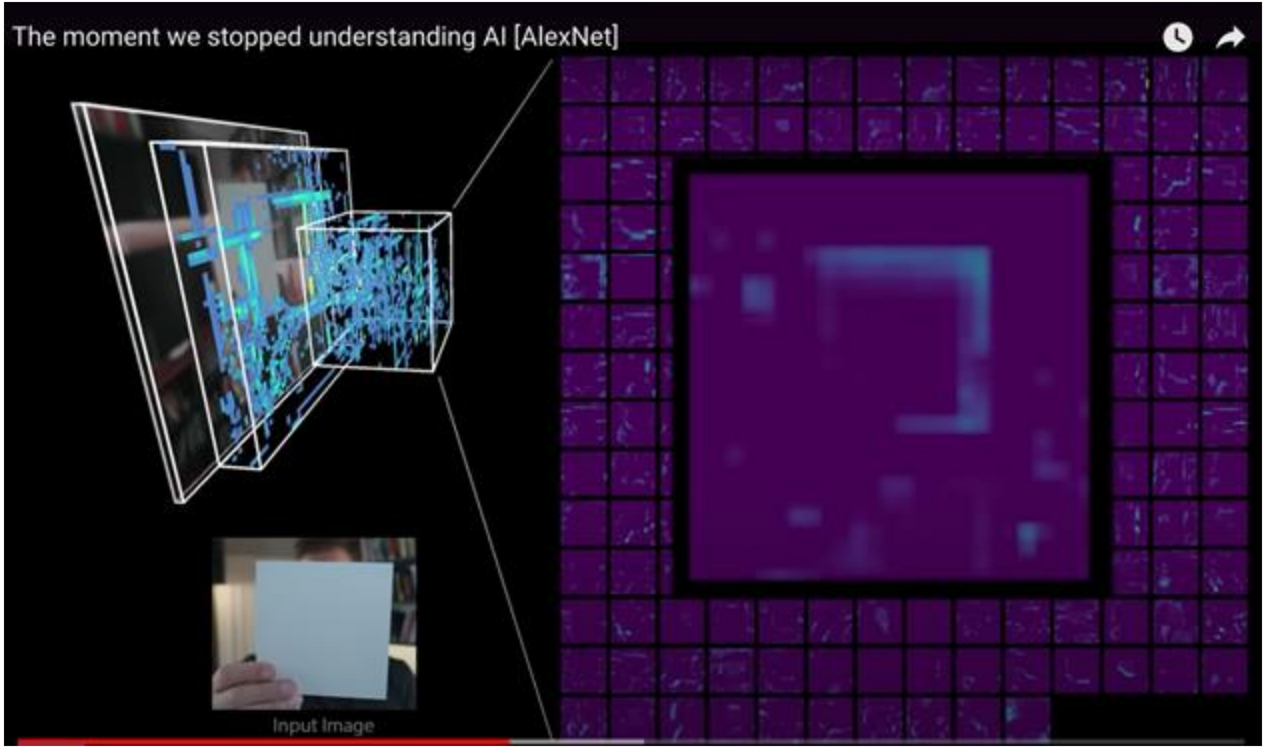
https://www.youtube.com/watch?v=UZDiGooFs54

Repeat convolution for deeper layers
But hard to visualize: 1 image becomes a stack of 96



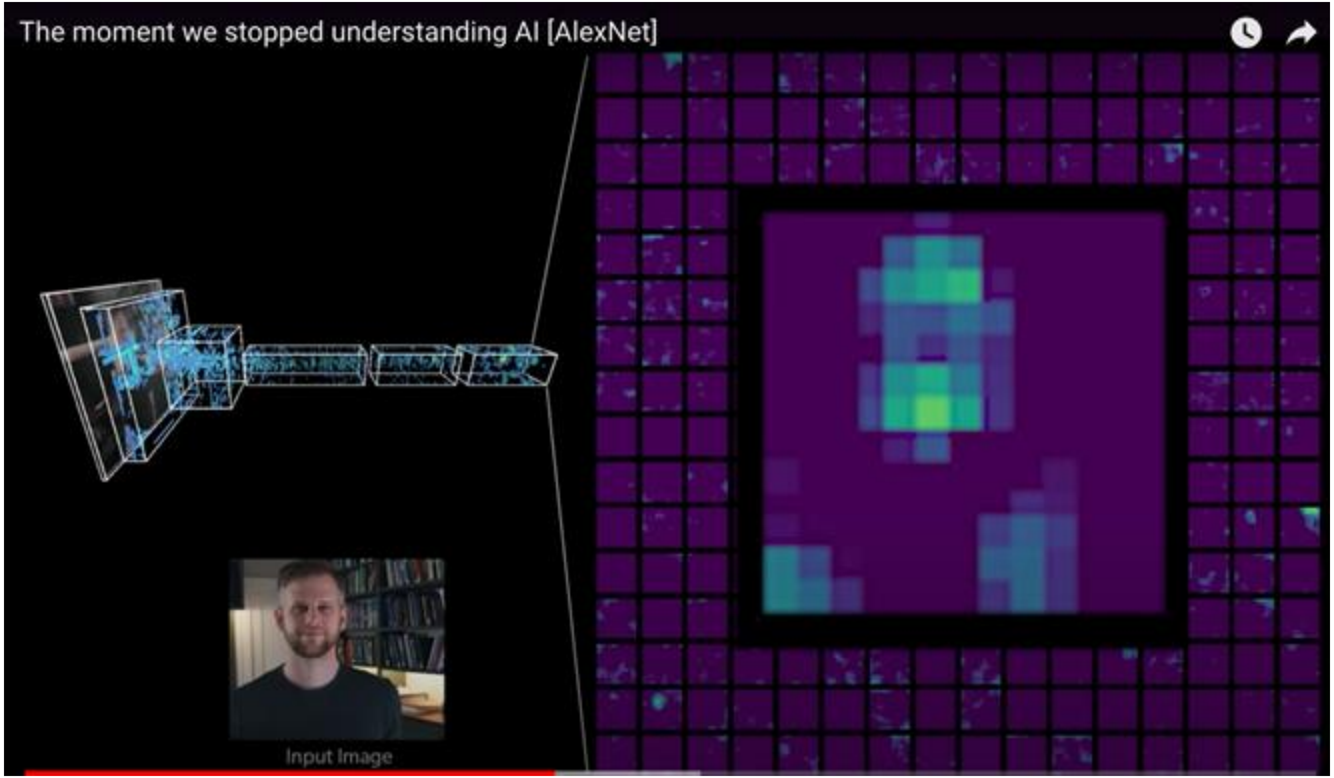
https://www.youtube.com/watch?v=UZDiGooFs54

Testing images shows the deeper layers are learning more complicated features



https://www.youtube.com/watch?v=UZDiGooFs54

Deeper layers recognize very complex features: faces!

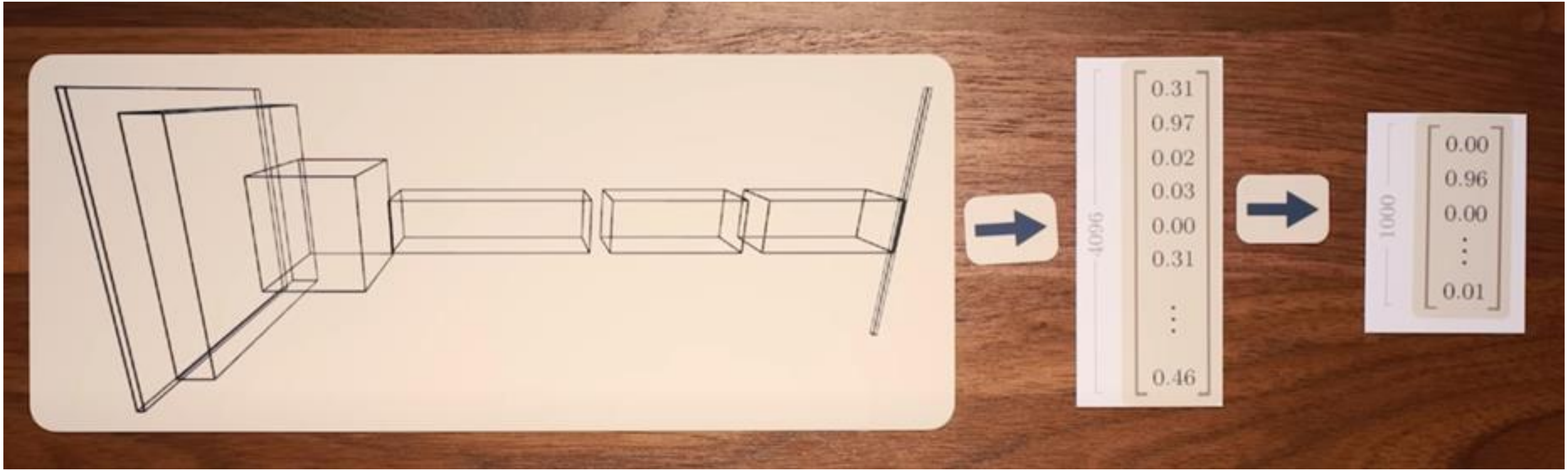


https://www.youtube.com/watch?v=UZDiGooFs54

Final two layers are special:

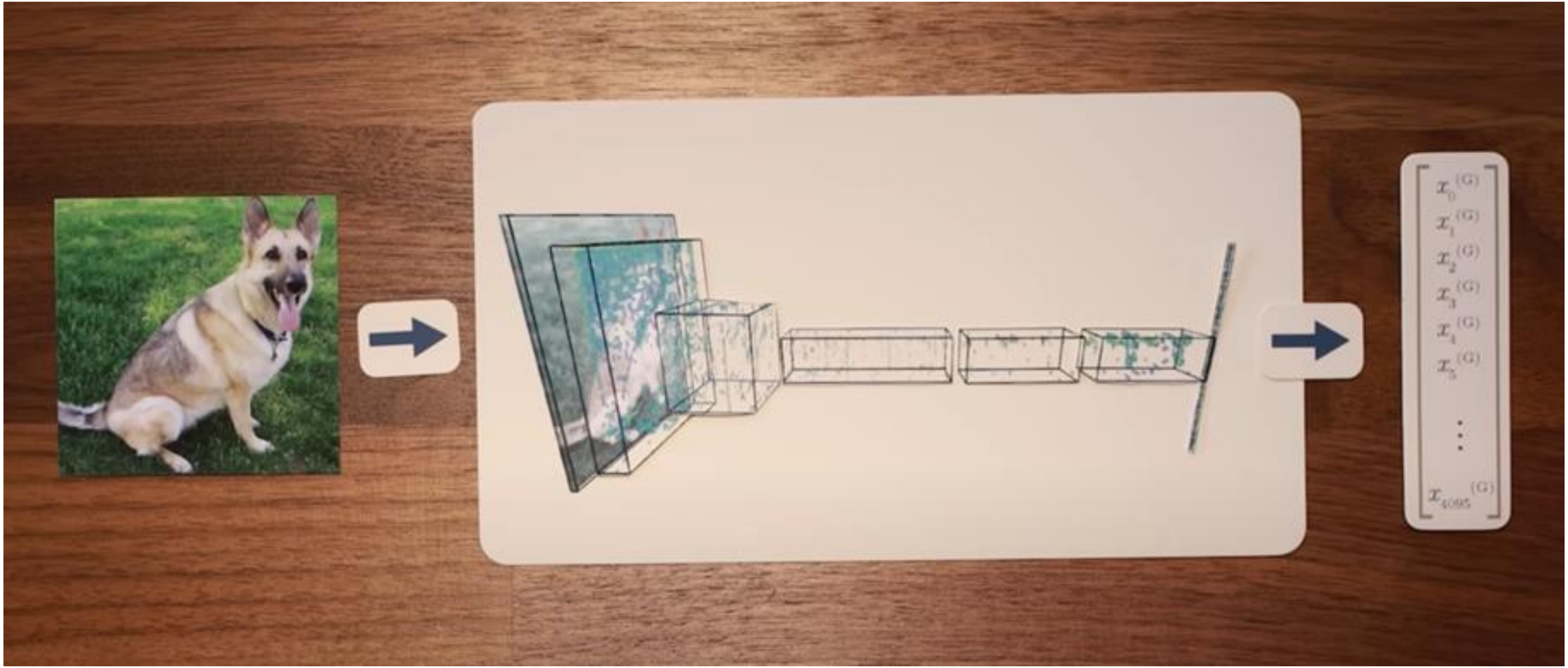
Last layer [1x1000]: Probability of the image in class i

Second last [1x4096]: Projection of image into 4096-dim space



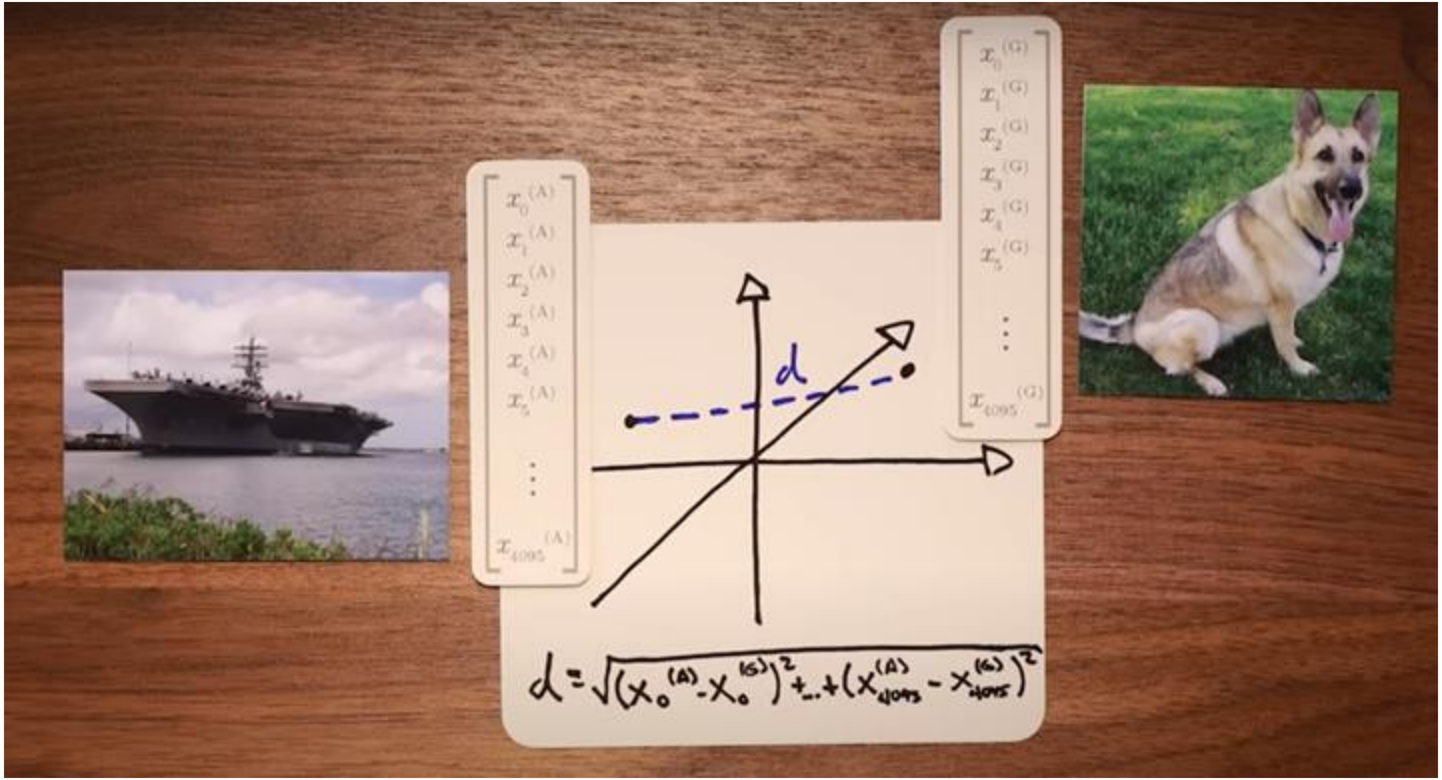
https://www.youtube.com/watch?v=UZDiGooFs54

Second to last forms an embedding of the input image



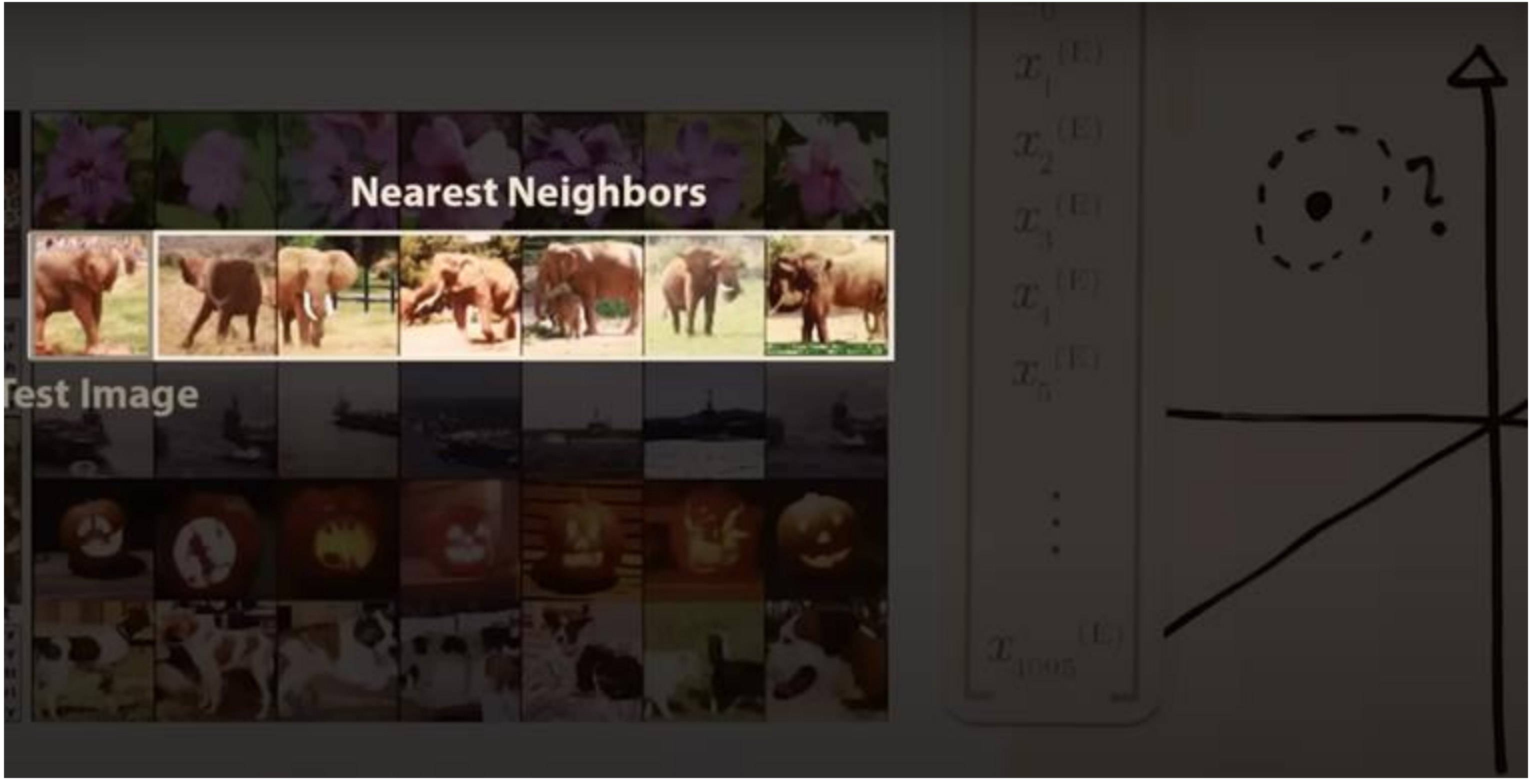
https://www.youtube.com/watch?v=UZDiGooFs54

Each image creates a unique vector
Compute the distance between vectors in 4096-dim space



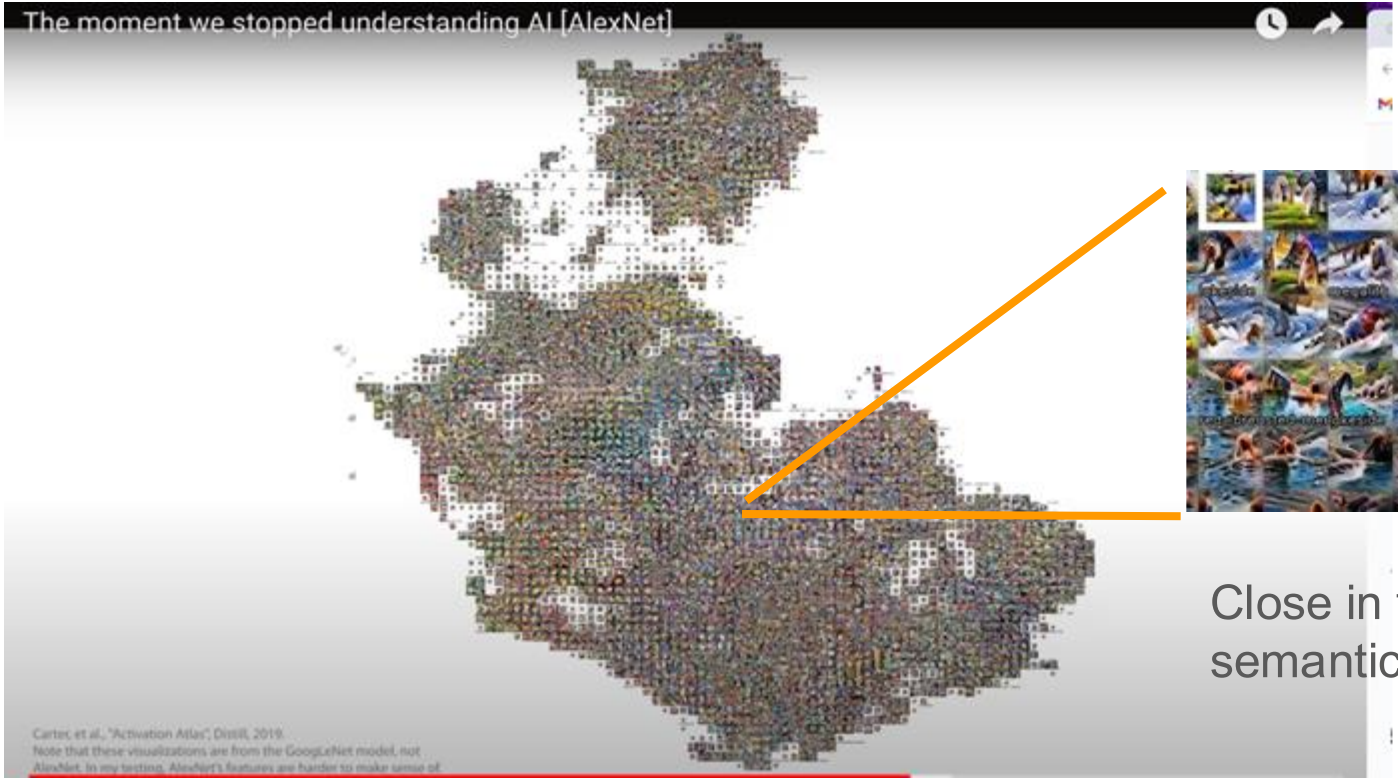
https://www.youtube.com/watch?v=UZDiGooFs54

Nearest neighbors in 4096-dim space are semantically related!
Robust to object orientation, size, color, etc



https://www.youtube.com/watch?v=UZDiGooFs54

Activation Atlas: tSNE plot of images embeddings



Close in the projection are semantically related

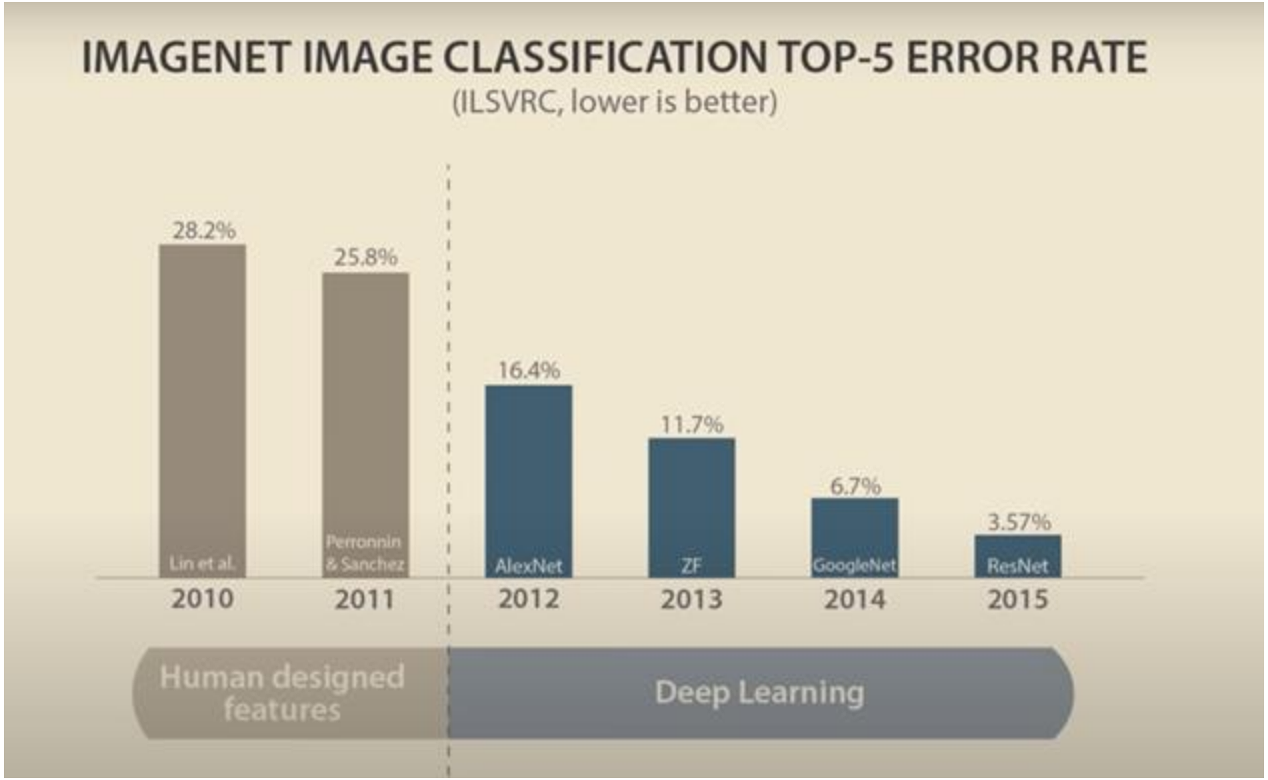
https://www.youtube.com/watch?v=UZDiGooFs54

Activation Atlas: tSNE plot of images embeddings

Directionality also caries meaning

https://www.youtube.com/watch?v=UZDiGooFs54

Deep Learning Success!



"Our results show that a large, deep convolutional neural network is capable of achieving record-breaking results on a highly challenging dataset using purely supervised learning. It is notable that our network's performance degrades if a single convolutional layer is removed. For example, removing any of the middle layers results in a loss of about 2% for the top-1 performance of the network. ***So the depth really is important for achieving our results***."

https://www.youtube.com/watch?v=UZDiGooFs54

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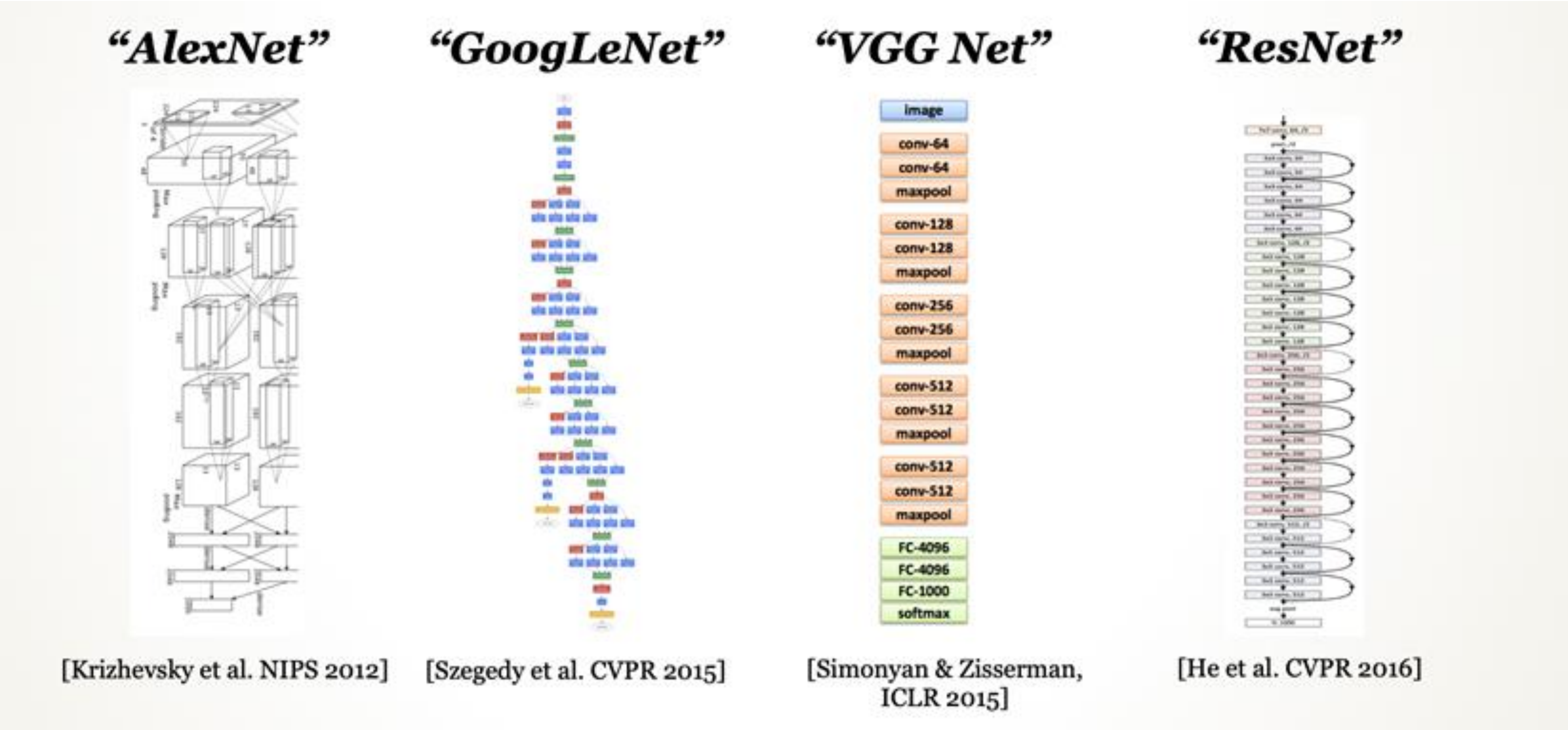
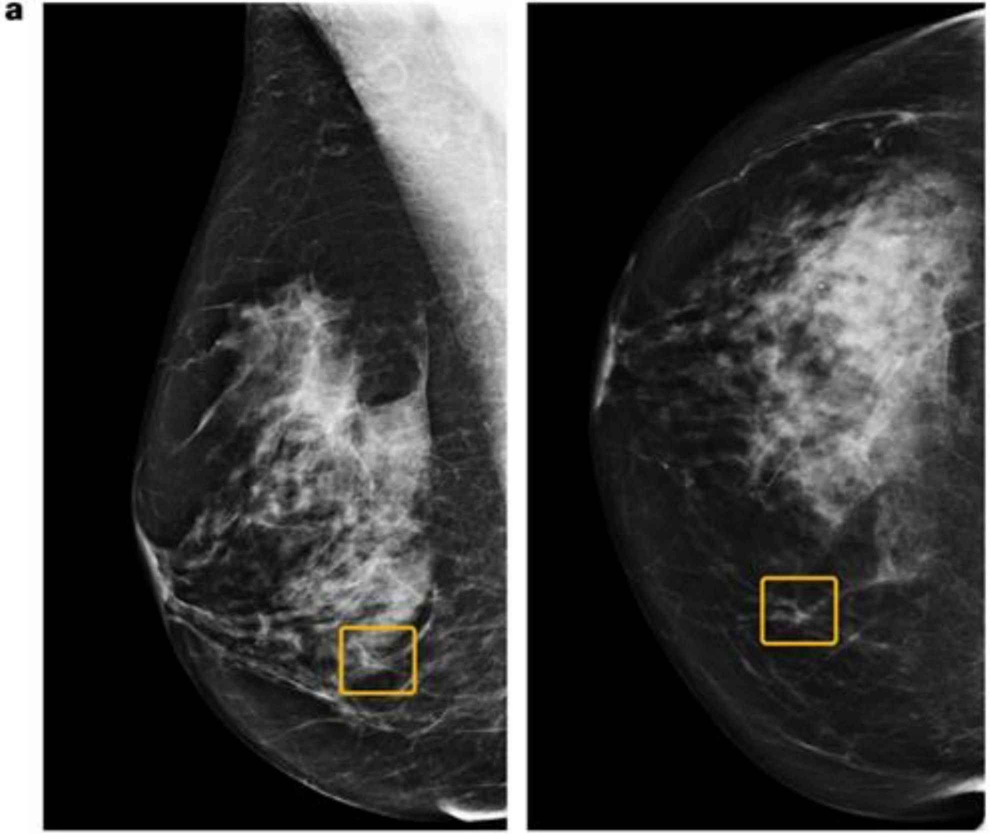
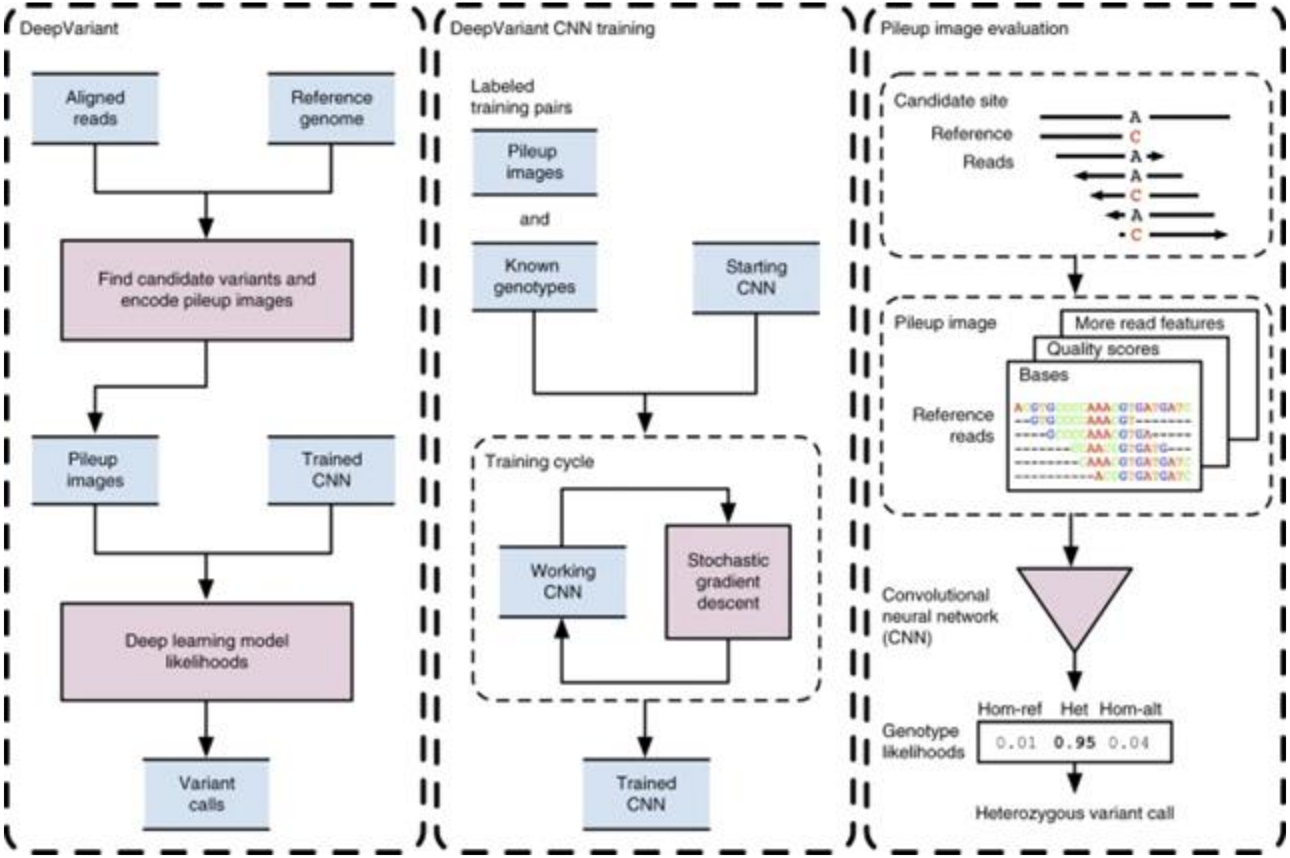


imagenet_ilsvrc2017_v1.0

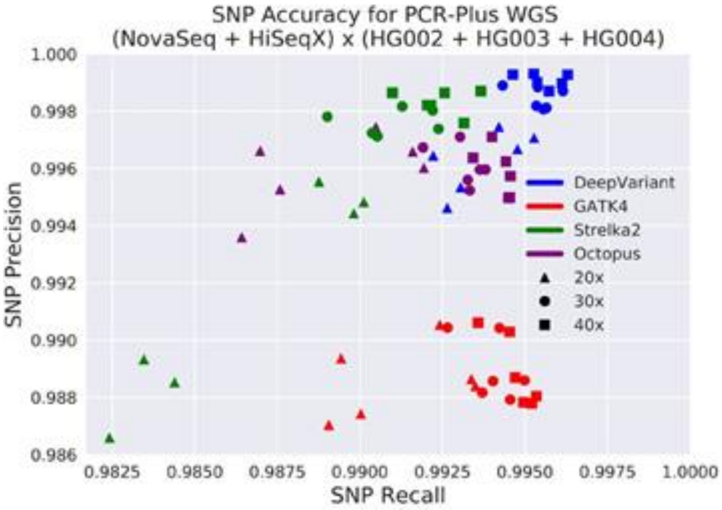


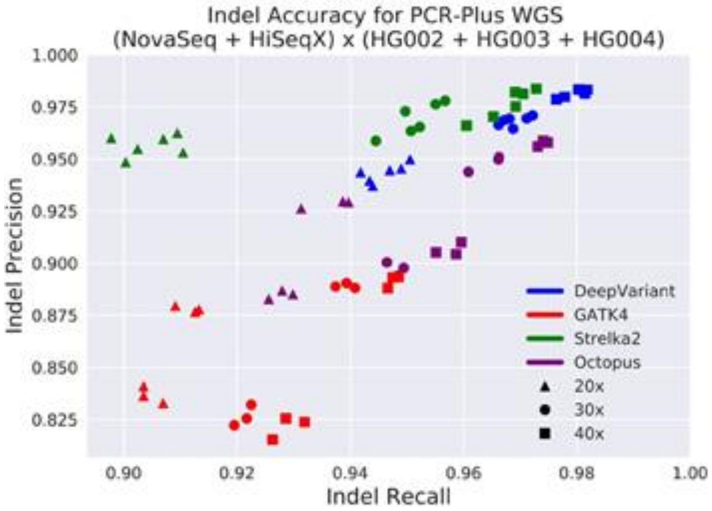
International evaluation of an AI system for breast cancer screening
McKinney et al (2020) Nature. https://doi.org/10.1038/s41586-019-1799-6

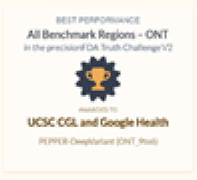


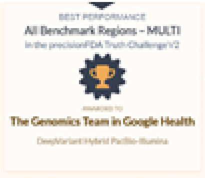
A universal SNP and small-indel variant caller using deep neural networks
Poplin et al (2018) Nature Biotechnology. https://doi.org/10.1038/nbt.4235

precisionFDA Challenges





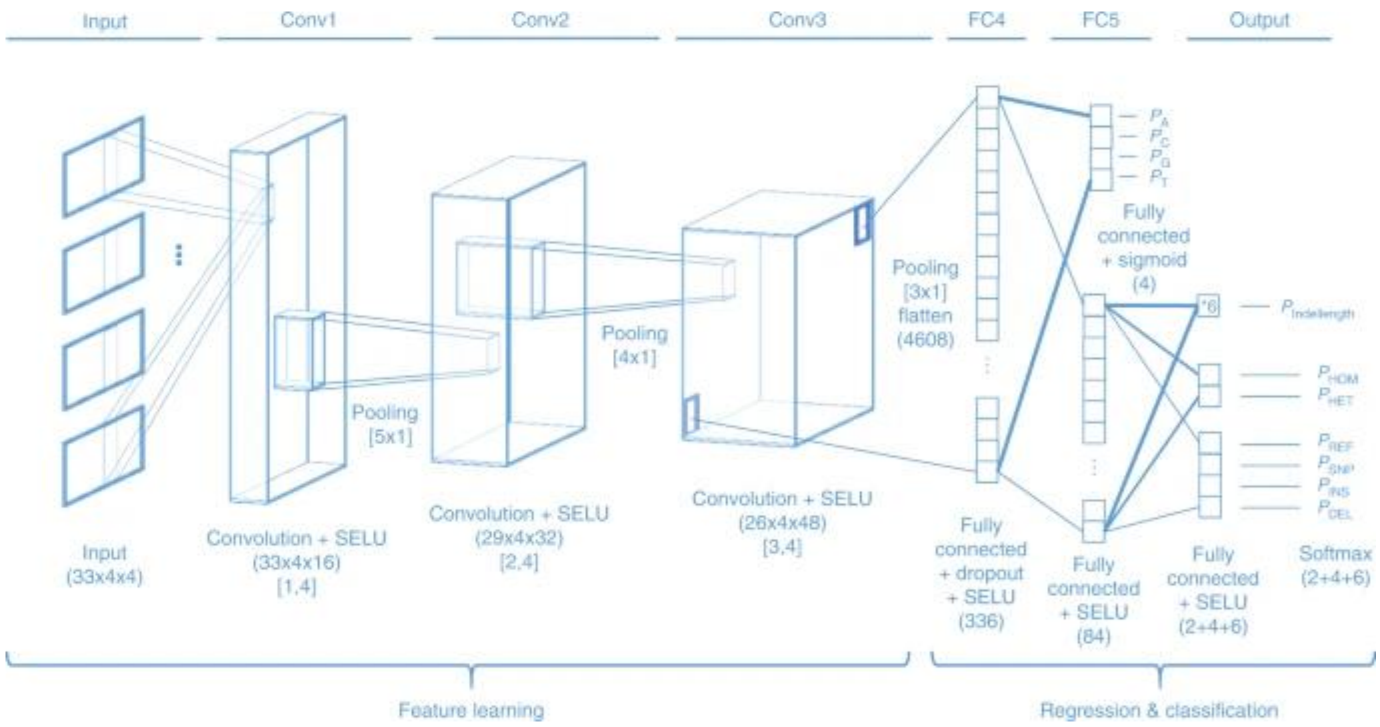




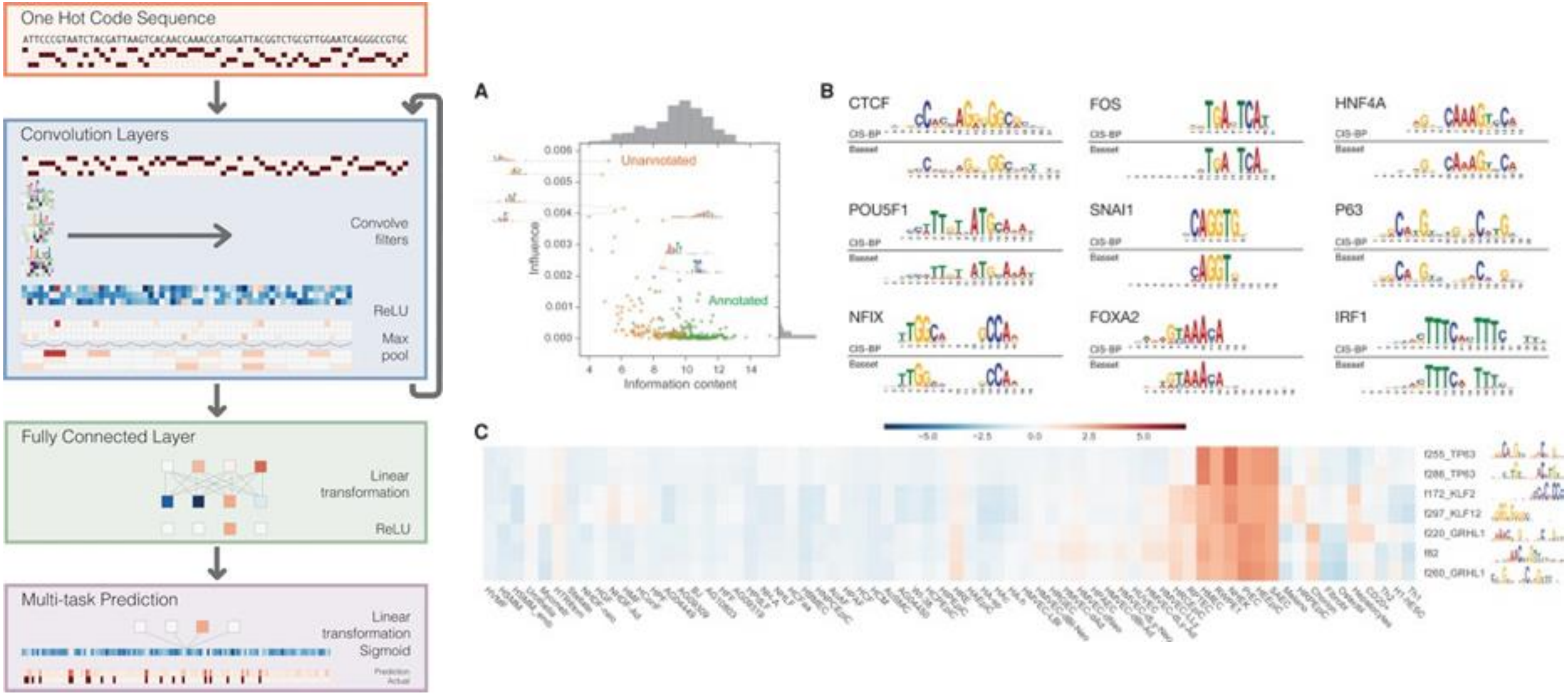
An Extensive Sequence Dataset of Gold-Standard Samples for Benchmarking and Development

Baid et al (2020) bioRxiv

https://www.biorxiv.org/content/10.1101/2020.12.11.422022v1.full



A multi-task convolutional deep neural network for variant calling in single molecule sequencing
Luo, Sedlazeck, Lam, Schatz (2019) Nature Communication.
https://doi.org/10.1038/s41467-019-09025-z



Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks
Kelly et al (2016) Genome Research. doi: 10.1101/gr.200535.115

ANNs are “Universal Approximators”

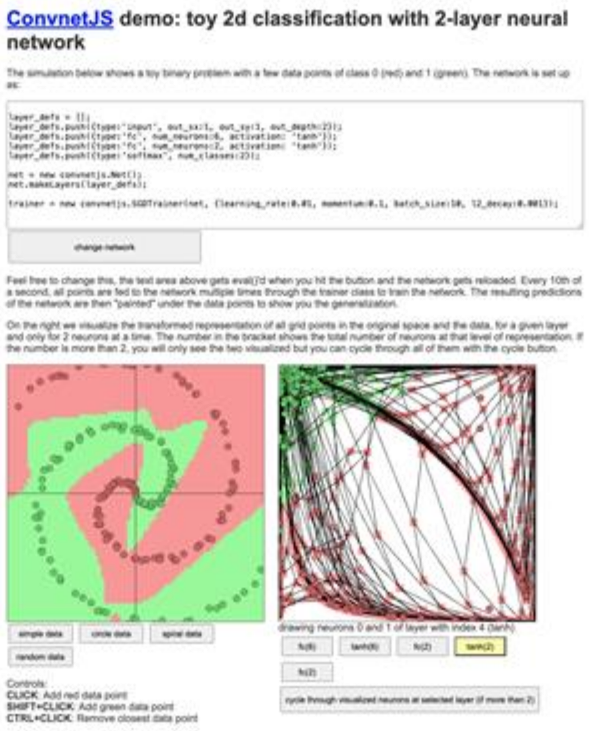
f() => 42

224*224 = 50k pixels

50k pixels * 3 bytes (RGB)
838B possible inputs

ANNs are “Universal Approximators”

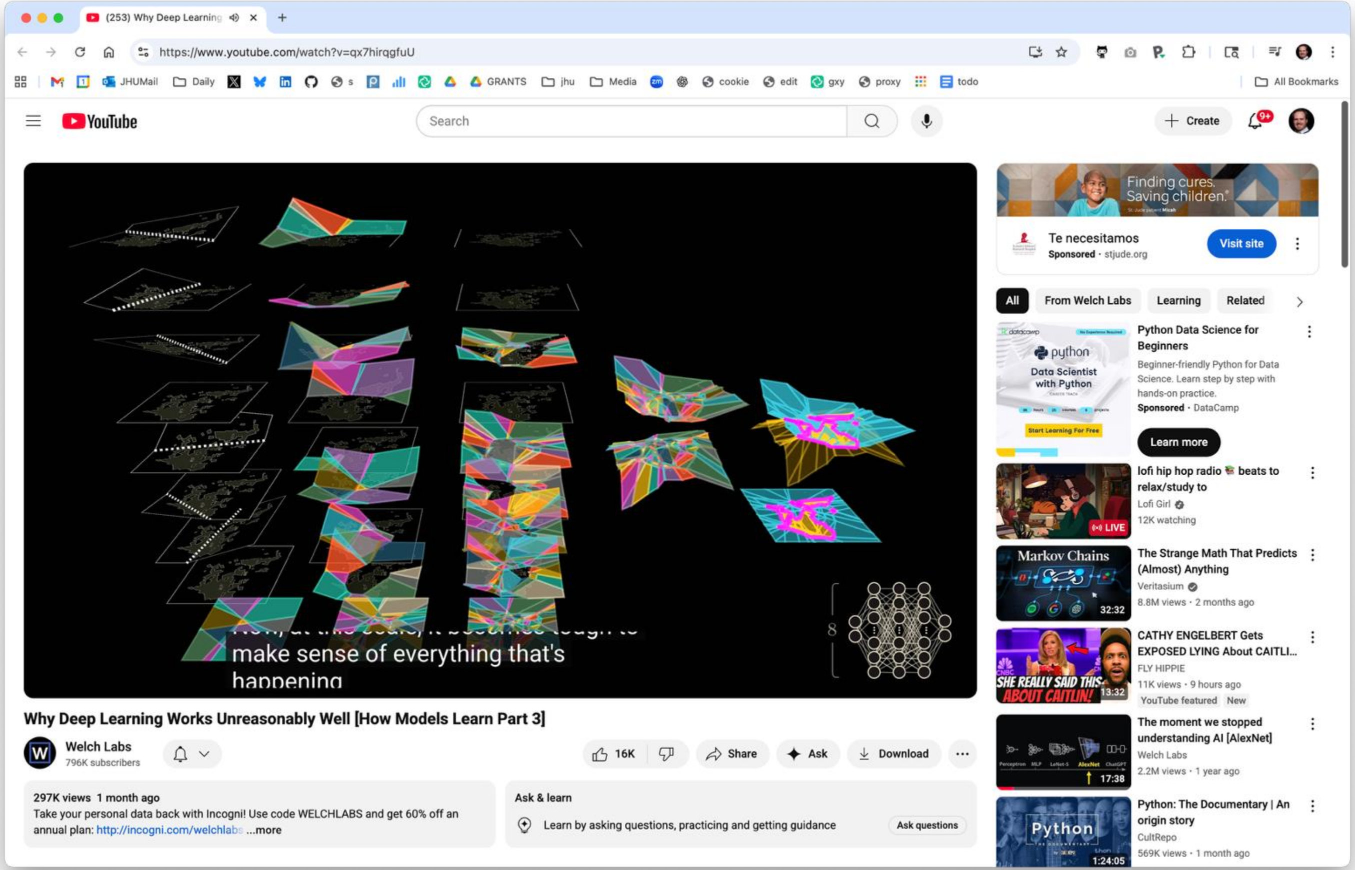
f() => 42



224*224 = 50k pixels

50k pixels * 3 bytes (RGB)
838B possible inputs

https://cs.stanford.edu/people/karpathy/convnetjs/demo/classify2d.html



Welch Labs: Why Deep Learning Works Unreasonably Well

https://www.youtube.com/watch?v=qx7hirqgfuU

Reflections

- **CNNs are incredibly powerful for image recognition (and similar tasks)**
  - Heavily inspired by how the human brain processes visual information from the real world
  - Start with the most basic feature kernels and then iterate into higher level concepts leveraging their spatial proximity
  - Ultimately projects images into abstract semantic embeddings

- **Certain problems are a great fit for CNNs:**
  - ImageNet: Lots of test data, nicely annotated into 1000 categories
  - DeepVariant/Clairvoyante: Lots of test data; One network architecture made it the worlds best variant caller for all types of long read data: PacBio, ONT, 10x

- **Deeper network architectures generally perform better**
  - Largely constrained by GPU RAM & Cores

“Images” in Bioinformatics

- **DNA sequence: one-hot encode**
  - Are there other encoding schemes? (modDotPlot)
  - Predict features (genes, repeat classes); Predict species (kraken)
- **Coverage tracks:**
  - Predict: SNVs (DeepVariant), CNVs, SVs (Cue)
  - RNAseq, ChIPseq (overcome error, allele specific)
- **Functional Genomics**
  - Inputs
    - Expression from multiple tissues (GTEx) from one patient
    - Multiomics: Variants + expression + methylation
  - Predict phenotype: Gender (easy), cell types (easy), age (pretty easy)
    - Phenopredict: https://academic.oup.com/nar/article/46/9/e54/4920847
  - Disease outcome: Needs the right training data, right tissues
- **Multiple alignments, position weight matrix**
  - Predict conserved regions, predict binding sites, etc

	LeNet-5	AlexNet	GPT-3	GPT-4
YEAR	1998	2012	2020	2023
TRAINING DATA	· MNIST Dataset · 60k training examples · 10 classes	· ImageNet Dataset (ILSVRC) · 1.2M training examples · 1000 classes	· Common Crawl, WebText, Wikipedia, others · ~500B training tokens · ~100k unique tokens	· ~13T training tokens*
TRAINING COMPUTE	· Pentium II CPU · ~0.27 GFLOPs	· Dual Nvidia GTX 580 · 3162 GFLOPs	· 10,000 Nvidia V100 GPUs · 1+ ExaFLOPs	· 25,000 Nvidia A100s GPUs* · ~4+ ExaFLOPs
ALGORITHM	· ~60k Parameters · 5 Layers · Sigmoid Activation Function	· ~60M Parameters · 8 Layers · ReLU Activation Function · Dropout	· 175B Parameters · 96 Layers · Transformers	· 1T+ Parameters* · *120 Layers · Transformers

https://www.youtube.com/watch?v=UZDiGooFs54

Exam 1 Review

What to study?

Closed book / no electronics in class exam [75 min]:

- One sheet of paper for notes (double sided)
- Recommend powerpoint / gslides / keynote but hand-written is okay
- Bring a pen / pencil!

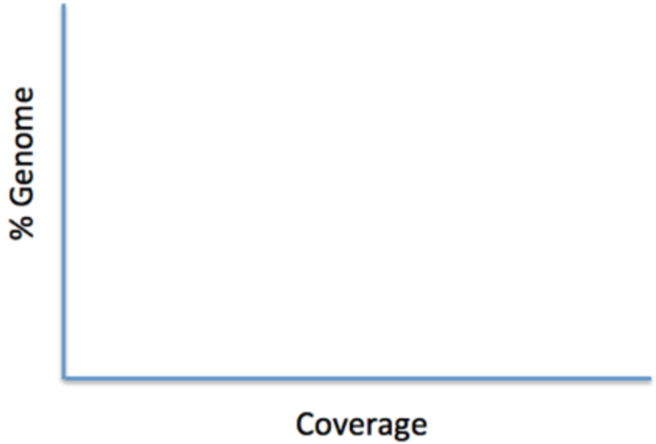
1. **Homework assignments**
   - Make sure you understand the key concepts, algorithms, data structures, and methods involved
   - Don’t worry about running software or writing code, but you should know what are the major software tools and how they work

2. **Lectures [with videos on canvas!]**
   - Make sure you understand the key concepts, algorithms, data structures, and methods involved :)
   - No “complicated” numerical computing, but you should be able to sketch a distribution, approximate values, & other simple calculations

3. **Readings**
   - These are to provide more details and provide additional perspectives
   - You are not responsible for plots/content only presented in the readings
   - Of course, you are free to read/watch/discuss additional resources

Sample Question

Q3. The Maryland blue crab genome is 1 Gbp in size. Approximately how many 100bp reads should we sequence so that we expect at least 99.85% of the genome will be sequenced at least 40 times? Sketch the expected coverage distribution for this number of reads; be sure to clearly label the mean coverage, and how 40 fold coverage relates to the mean. (Hint: In a normal distribution, 68.2% of the data is within 1 standard deviation of the mean, 95.4% within 2, 99.7% within 3, and 99.9% within 4)



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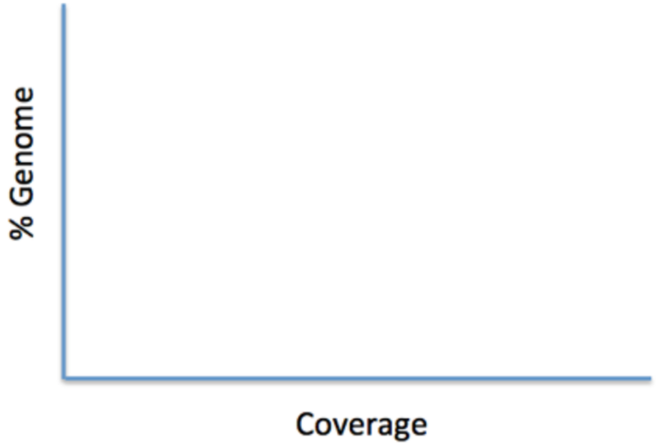

Also expect a few multiple choice, short answer, and longer essay questions

Exam Topics

Genomics

- Genomics Technologies
  - Illumina, PacBio, Nanopore
- Coverage, Kmers, Motifs
- Genome Assembly
- Repeats, Heterozygosity
- Whole Genome Alignment
- Dotplots
- Jaccard Coefficient
- Average Nucleotide Identity
- Read mapping
- Variant Identification

Quantitative Techniques

- Normal, Poisson, Binomial
- P-value
- de Bruijn and overlap graphs
- Quality Values (Phred Scale)
- Full text indexing
- BWT & Suffix Arrays
- Seed & Extend
- Edit Distance
- PCA / t-SNE / UMAP

What is the goal? What is the approach? What are the key challenges?

How did we explore these topics in the homeworks and lectures?